O4e - Focal Length and Principal Planes of a Lens System

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Abstract

This report investigates the optical behavior of a two-converging lens system, f_{sys} determined through mathematical modeling, Bessel's Formula and lens equations. Results include $f_{sys}^{g'} = (0.178 \pm 0.00813)$m and $f_{sys}^{b'} = (0.173 \pm 0.00725)$m (With $r^2 = 0.98366$ and $r^2 = 0.98608$ respectively), and a Bessel-derived $f_{sys}^{B} = (0.18438 \pm 0.00055)$m. Additionally, focal lengths $f_1 = (0.10229 \pm 0.00030)$m and $f_2 = (0.32054 \pm 0.00018)$m were determined for each lens. Bessel's Method proved most accurate despite challenges in determining accurate image planes. Calculated image plane distances $h = (-0.01638 \pm 0.01097)$m and $h' = (-0.26597 \pm 0.04012)$m exhibited unexpected signs, likely due to unrecorded signs of magnification. The Ray Diagram constructed showed a theoretical magnification M^D = -2.49 diverging from experiment $M^{Exp} = 2.39 \pm 0.0641$ with 5.22% adjusted uncertainty between the two. Overall the results align with theoretical predictions with deviations resulting from experimental error. For future experimentation it is recommended to record the signed magnification, consider spatial uncertainty, and increase data collection within a reasonable magnification range.

1 Introduction

Optical lenses and Optics play a big part in the modern world, from phone cameras, to medical equipment to satellite communication and imagery [3]. This report will explore the optical behavior of a compound lens system consisting of two converging lenses, and compare two methods for determining the focal length of the system. Along with exploring other factors such as magnification and the placement of the system's image planes. This report will utilize techniques within the field of Geometric Optics which takes the core assumptions that [3]: Light travels in a straight line and behaves like a ray, and only refraction and reflection of light occurs, *No scattering, absorption, diffraction, etc.*
The first method involves measuring the magnification M of the object through the two lens system below.

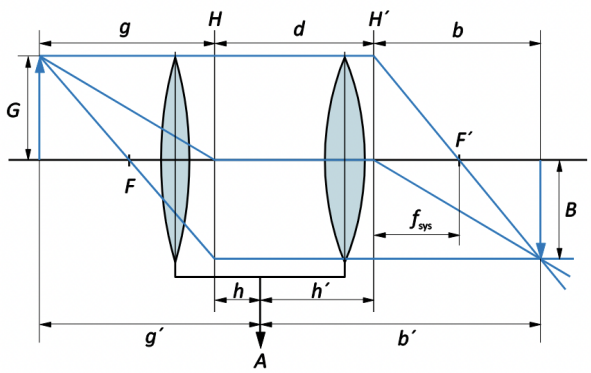

Figure 1: Task I Optical Diagram

Source: [4]

and along with distance measurements the following two imaging equations.

$$g' = g + h = f_{sys}(1 + \frac{1}{M}) + h \tag{1}$$

$$b' = b + h' = f_{sys}(1 + M) + h' \tag{2}$$

Where: [1]

$$M = \frac{B}{G} = -\frac{b'}{g'}$$

is essential in determining the focal length of the system.f_{sys}.
The second method utilizes the reciprocal nature of the magnification to record the two positions, g_1 and g_2 at which a lens can focus an image on a static screen at s, as seen below.

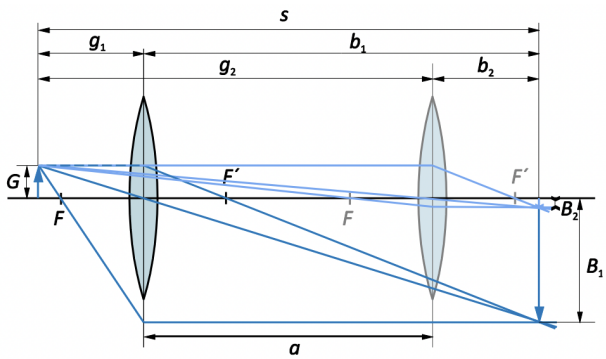

Figure 2: Task II Optical Diagram

Source: [4]

Then through the use of Bessel's Formula:

$$f = \frac{s^2 - a^2}{4s} \tag{3}$$

The focal length of one of the lenses can be found. by repeating this process with the other lens the f_{sys} can be found with:

$$\frac{1}{f_{sys}} = \frac{1}{f_1} + \frac{1}{f_2} \tag{4}$$

2 Task I

2.1 Experimental Setup

Through use of a linear rail, Task I was setup with a lamp and an accompanying, illuminated, 1mm scale attached

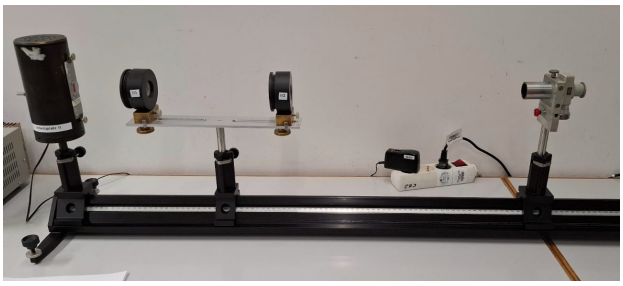

Figure 3: Task I experimental setup

to the far left of the rail as seen in Fig 3. Attached to the right of the lamp is a two-lens system with a fixed separation distance of 245mm and finally an ocular with an internal 0.1mm scale.

Over the course of 10 different measured distances between the illuminated scale and the center of the two-lens system. The ocular was positioned in the image plane of the illuminated scale and the image's magnification was measured by measuring the separation distance between markings in the 1mm scale with the 0.1mm scale in the Ocular. Additionally the distance from the center of the two-lens system and the Ocular was measured.

Then with the measurements obtained, a linear regression was performed through use of Eq. 1 and Eq. 2 to determine an experimental value for f_{sys}, along with distances to the image planes of H and H' respectively, (Fig.1).

2.2 Data Analysis

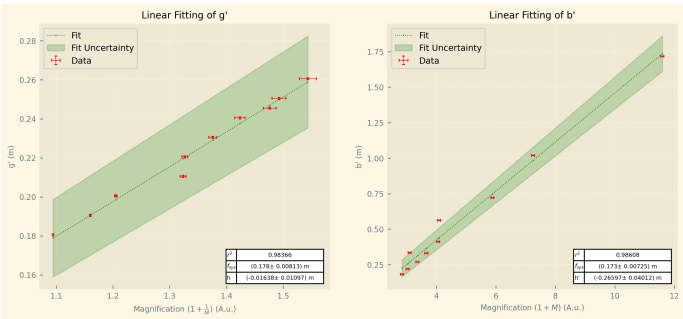

Figure 4: Linear Regression of a) g' and b) b'

As shown in Fig. 4 the determined value for f_{sys} from a linear regression of Eq. 1 and Eq. 2 are $f_{sys}^{g'} = (0.178 \pm 0.00813)$m and $f_{sys}^{b'} = (0.173 \pm 0.00725)$m with accompanying h =(-0.01638 ± 0.01097)m and h' =(-0.26597 ± 0.04012)m respectively. Along with these results the g' linear regression has a r^2 =0.98366 and b' a r^2 =0.98608 which both suggest a strong correlation with determined parameters depicting a linear trend that of the distances g' and b' increasing as $1 + \frac{1}{M}$ and $1 + M$ increase respectively.
It is peculiar that both values h and h' are negative as these values depict a distance from the center of the two-lens system on either side, thus it is expected that either both h and h' be positive to signify their respective distances from A as depicted in Fig. 1. Or one be positive and the other negative, in the case that Eqs. 1 and 2 returned signed distances about the center point of the lens system. However as neither is the case it is predicted that failure to collect signed values for the magnitude affected the results of h and h'. Thus to relate these values to physical meaning; $h = (-0.01638 \pm 0.01097)$m has been interpreted as being behind the center point A, Fig.1, and $h' = (-0.26597 \pm 0.04012)$m as being in front of A.

It was noticed during experimentation that most images viewed through the Ocular were inverted, however due to a lack of foresight and difficulty in determining whether some images were inverted, due to a high magnification, the signed magnification was not measured. However it is predicted that this does not impact the results of $f_{sys}^{g'}$ and $f_{sys}^{b'}$ as these values represent the gradient and would only result in a change of sign.

The uncertainty in $f_{sys}^{g} = (0.178 \pm 0.00813)$m and $f_{sys}^{b'} = (0.173 \pm 0.00725)$m is moderate, conveying a moderate degree of precision. However both f_{sys} measurements fall within the uncertainty of each others, suggesting a degree of accuracy in the method as a whole. The degree of accuracy in relation to this experiment is predicted to be less than that of expected in account of failure to collect signed magnification values.

3 Task II

3.1 Experimental Setup

For the second task, attached to the linear rail is a lamp that illuminates an object, one of the lenses from Task I and a screen placed beyond the lens that was used to find the image plane. Initially the lens was positioned close to the illuminated object and the distance between the two measured. The screen was then moved until a sharp, upright image was formed and the distance between lens and the screen was recorded. The lens was then varied until a sharp, inverted image was displayed on the screen and the distance between the two measured. This process was repeated a total of three times by varying the lens and screen until a unique set of upright and inverted images were found.

The lens was then replaced by the second lens used in Task I and the process repeated. After all measurements were obtained Eq. 3 was used to determine the focal length for each lens with $a = b_1 - b_2$ and $s = g_1 + b_1 = g_2 + b_2$, f_1, f_2, Fig. 2. Finally f_1 and f_2 were used in Eq. 4 to determine focal length f_{sys} of the system used in Task I via Bessel's Method. This value was subsequently compared to the results of Task I.

3.2 Data Analysis

Lens I f_1 **(m)**	**Lens II** f_2 **(m)**
0.10214 ± 0.00043	0.31211 ± 0.00024
0.10236 ± 0.00043	0.33658 ± 0.00023
0.10236 ± 0.00043	0.31293 ± 0.00026
Avg: 0.10229 ± 0.00030	**Avg:** 0.32054 ± 0.00018
$f_{sys} = 0.18438 \pm 0.00055$ **m**	

Table 1: Focal Length Measurements

The results of Table. 1 show that a consistent result 0.102 m for f_1 with all variation in the the results falling within

the uncertainty of each other's result, suggesting high precision of measurements. The results of f_2 differ by a significant degree between recordings, with the first recording (0.31211 ± 0.00024)m and the third, (0.31293 ± 0.00026)m are relatively close but still distinct and the second result of (0.33658 ± 0.00023)m significantly larger and is suspected that this result is due to human error during the experimental process however as no discernible reason can be explained as to why this value has such a discrepancy from the other measurements it remained a valid data point.

The determined value for f_{sys}^{B} via the Bessel's formula and Eq.4 is $f_{sys}^{B} = (0.18438 \pm 0.00055)$m which falls within the uncertainty of the experimental result $f_{sys}^{g'} = (0.178 \pm 0.00813)$m but outside of the result $f_{sys}^{b'} = (0.173 \pm 0.00725)$m from Task I. Despite this, Bessel's method of determining f_{sys} of the focal system in Task I is still considered to be accurate and precise within the field of geometric optics and when considering experimental error.

Whilst collecting measurements, there was some difficulty in determining the exact location of the image place to set the screen at, particularly with the upright image which was consistently much larger than the area of the screen, However this was not the case for the f_2 result (0.33658 ± 0.00023)m and it is unknown as to why this result deviates significantly from the remaining data.

4 Task III

4.1 Data Analysis

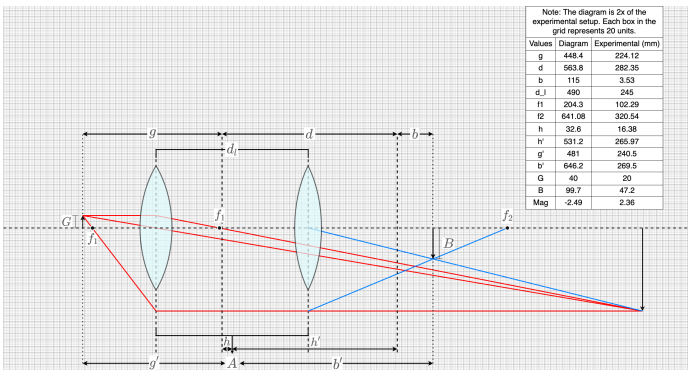

Figure 5: Ray Diagram of Task I setup

Through Ray Optic techniques, and use of a data point collected in Task I, along with the focal lengths obtained in Task II. Values for the variables seen in Fig. 5 and Table 2. was determined, and in particular a value for the linear Magnification. the value obtained for $M^D = -2.49$ shows that indeed for this data point the linear magnification is negative and thus the image was inverted.

Percentage Uncertainty in Linear Magnification:

$$\left| \frac{|M^{Exp}| - |M^D|}{|M^D|} \right| \times 100 \tag{5}$$

$$\left| \frac{2.36 - 2.49}{2.49} \right| \times 100 = 5.22\% \tag{6}$$

For this calculation absolute values of both the magnification were used, since during the experiment the linear magnification was not measured with its sign even though the image observed through the ocular was inverted.

Values	$\frac{1}{2}$Diagram (m)	Experimental (m)
g	0.2242	0.22412 ± 0.0005
d	0.2819	0.28235 ± 0.0005
b	0.0575	0.00353 ± 0.0005
d_l	0.245	0.245
f_1	0.10215	0.10229 ± 0.0005
f_2	0.32054	0.32054 ± 0.0005
h	0.0163	0.01638 ± 0.0005
h'	0.2656	0.26597 ± 0.0005
g'	0.2405	0.24050 ± 0.0005
b'	0.3231	0.26950 ± 0.0005
G	0.020	0.02000 ± 0.0005
B	0.04985	0.04720 ± 0.0005
Mag	-2.49	2.36 ± 0.0641

Table 2: Diagram vs. experimental measurements

From Table. 2 a direct comparison between $M^D = -2.49$ and M^{Exp} =2.39 ± 0.0641 shows that the absolute value of the linear magnifications are still distinct and the uncertainty in M^{Exp} does not encompass M^D. This is most likely a result of inaccuracies in constructing a scaled diagram, and difficulty in determining the exact location of the image plane which is further supported when comparing the Diagram and Experimental values of b in Table. 2, of $b^{Exp} = 0.00353 \pm 0.0005$ and $b^D = 0.0575$ which are vastly different. *Naturally this difference also extends to* b', it is uncertain whether more factors are involved in this discrepancy.

5 Discussion

5.1 Task I & Task II

When comparing the results of Task I, $f_{sys}^{g'} = (0.178 \pm 0.00813)$m and $f_{sys}^{b'} = (0.173 \pm 0.00725)$m respectively, with Task II $f_{sys}^{B} = (0.18438 \pm 0.00055)$m. The value of f_{sys}^{B} and its uncertainty both fall within the experimental error of $f_{sys}^{g'}$ but the $f_{sys}^{b'}$ is distinct suggesting some accuracy between the methods used. Additionally the uncertainty in f_{sys}^{B} is significantly lower than that of $f_{sys}^{g'}$ and $f_{sys}^{b'}$ possibly conveying a higher degree of precision when employing Bessel's Method. It should also be noted that both methods encountered difficulties in determining the image plane through use of the Ocular or Screen respectively, however Task II proved to have more difficulty with this factor, particularly for the case of the upright image. The measurements for Task I also did not record the the sign of the magnitude for each value obtained and although it is not expected to impact $f_{sys}^{g'}$ and $f_{sys}^{b'}$ this may be the case. It is also believed that there was an unaccounted error in Task I relating to the use of three lenses to measure the magnification and factors relating to the scale of the discernible markings on the object.

The results of h and h' as determined in Fig. 4 of $h = (-0.01638 \pm 0.01097)$m and $h' = (-0.26597 \pm 0.04012)$m, proved confusing in their interpretation as from Eqs. 1 and 2 it is predicted that h and h' are absolute distances from the center point A ,Fig. 1, or h and h' are signed distances from A and in that case one is positive and one

negative thus the results obtained, being both negative, are expected to be misleading due to the failure to record signed magnification values for M. Thus the experimental values recorded where interpreted as the magnitudes of h being the distance behind the center point A and h' being the distance in front of A.

The results of Task II are predicted to more precise and accurate than that of Task I, in part to its simpler measurement method and that Task I involved mathematical modeling to derive two results for f_{sys}. Furthermore the process by which f_{sys}^{B} was determined was two straightforward calculations using Eq. 3 and 4 and experimentally involved less opportunities for measurement error, parallax error and focal error due to measurements involving only one lens at a time. Although this method is expected to be more accurate than that of Task I, it should be noted that the uncertainty recorded during experimentation was that of the systematic uncertainty in the linear rail and did not consider the spatial uncertainty in the location of the image plane, which is predicted to increase the uncertainty by a considerable amount.

For future determination of a lens system via Bessel's method it is recommended to record more sets of lens positions for the upright and inverted images and to account for the spatial uncertainty in the location of the image plane when recording measurements. Additionally it is recommended to take sets where the images are not significantly larger than the screen, particularly for the upright image. In regards to the process used in Task I, it is recommended to measure the signed magnification, collect more data points and focus on data points that do not involve a high magnification as uncertainty measurements increase rapidly with magnification.

5.2 Task III

The ray diagram was constructed using the values of f_1 and f_2 obtained from Task II, d_l and the position of A from data provided in the lab manual, the object distance was chosen from the collected data (0.2405 m) and a suitable object height was chosen in order for the object and image lengths to fit the diagram's scale. For assistance during the image construction, the ray optics tool from GeoGebra [2] was used. It can be noted that the resulting magnification from the ray diagram, M^D = -2.49 is close to experimental data with a percentage uncertainty of 5.22%, but does fall outside of the uncertainty of the $M^{Exp} = 2.39 \pm 0.0641$.

However, the discrepancy between the constructed value of $b' = 0.3231$ and the measured value (0.26950 ± 0.0005) is much more significant. this can be attributed to several contributing factors. First, inaccuracies in the scale drawing may have introduced errors, as the diagram is a geometrical construction where small deviations in ray tracing or scale selection can significantly affect the result. Furthermore, the object distance g' that was used to construct diagram was scaled and subsequently rounded to a suitable number and as such slightly differs from exact scale, resulting in a deviation from exact image distance.

In terms of experimental uncertainties, this includes slight misalignment of the lenses or Ocular, along with human error in deciding distances. Additionally, optical imperfections such as spherical or chromatic aberrations could have cause the image to appear sharp at an incorrect position. Finally if the Ocular's scale was not perfectly aligned, the image position may have been inaccurately recorded. Together, these factors account for the observed deviation of Diagram results against experimental values of b'.

6 Conclusion

This report investigated the optical behavior of a compound lens system consisting of two converging lenses. The focal length, f_{sys}, of the system was determined through use of mathematical modeling of Eqs.1, $f_{sys}^{g'} = (0.178 \pm 0.00813)$m and 2, $f_{sys}^{b'} = (0.173 \pm 0.00725)$m, and through Bessel's Equation Eq.3 to determine $f_1 = (0.10229 \pm 0.00030)$m and $f_2 = (0.32054 \pm 0.00018)$m and the lens equation Eq.4, $f_{sys}^{B} = (0.18438 \pm 0.00055)$m. The mathematical model for $f_{sys}^{g'}$ and $f_{sys}^{b'}$ both strongly agreed with experimental data with $r^2 = 0.98366$ and $r^2 = 0.98608$ respectively, the models were performed with the magnitude of M and not its signed value. It was determined that Bessel's Method was more accurate and precise despite difficulties in determining image planes.

Through the mathematical modeling process the image planes $h = (-0.01638 \pm 0.01097)$m and $h' = (-0.26597 \pm 0.04012)$m were determined. The sign of these values proved perplexing, not matching expected values, this is predicted to be a result of the failure to collect the sign of the magnification during experimentation.

A Ray Diagram was also constructed to compare the experimental magnification $M^{Exp} = 2.39 \pm 0.0641$ with theoretical values obtained through Ray Diagram Techniques. The diagram magnification M^D = -2.49, fell outside the experimental uncertainty of the M^{Exp}, however the percentage uncertainty of 5.22% lead to the conclusion that the deviation was due to the inaccuracies in scaling, alteration of object distance and accompanying experimental uncertainties.

Across all comparison and measurements, the experiment possessed an overall agreement with theoretical models with the deviations expected to be a result of experimental error. It is recommended in future to account for spatial uncertainty in the position of the image plane, record the signed magnification and collect more data points that are of a reasonable magnification.

References

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